

E-Commerce Infrastructure

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- Electronic commerce is built on top of a number of different technologies.
- These various technologies created a layered, integrated infrastructure that permits the development and deployment of electronic commerce applications.
- Each layer is founded on the layer below it and cannot function without it.

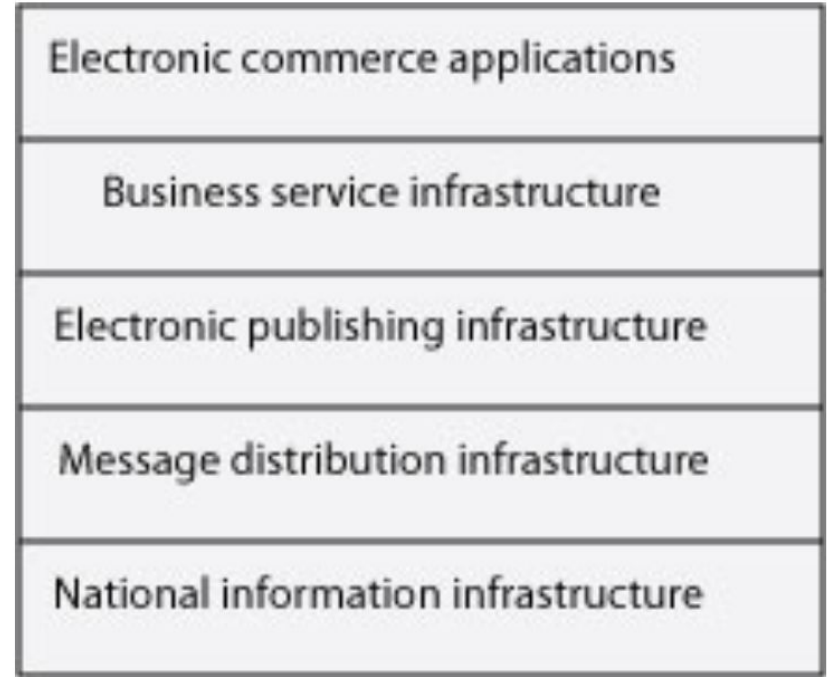


Fig: Electronic commerce infrastructure

National information infrastructure(NII)

- This layer is the bedrock of electronic commerce because all traffic must be transmitted by one or more of the communication networks comprising the national information infrastructure (NII).
- Whenever an online activity is performed, such as:
 - a. Placing an order on **Daraz**,
 - b. Sending money through **bKash**,
 - c. Making a payment on **Amazon**,
 - d. Sending an email using **Gmail**,

all of the data (traffic) travels through communication networks such as the **Internet, Wi-Fi, mobile networks, computer networks, and telephone networks**.

The collection of these communication networks is called the **National Information Infrastructure (NII)**.

That is why the **NII is called the bedrock (foundation) of electronic commerce**—because without these networks, no online transaction or communication can take place.

National information infrastructure(NII)

- The components of an NII include the
 - **TV and radio broadcast industries,**
 - **cable TV,**
 - **telephone networks,**
 - **cellular communication systems,**
 - **computer networks,** and
 - **the Internet.**
- The trend in many countries is to increase competition among the various elements of the NII to increase its overall efficiency because it is believed that an NII is critical to the creation of national wealth.

National information infrastructure(NII)

- **Example 1: Online Shopping**

A customer opens the **Daraz** app on a smartphone and orders a laptop.

- a. The **cellular network or Wi-Fi** connects the phone to the Internet.
- b. The **Internet** sends the order to Daraz's server.
- c. **Computer networks** process the order and update the inventory.
- d. The **telephone network** is used by the delivery agent to contact the customer.

This entire communication system is part of the National Information Infrastructure (NII).

National information infrastructure(NII)

- **Example 2: Online Banking**

A customer transfers money using a banking app.

- a. The smartphone connects through **Wi-Fi or a cellular network**.
- b. The **Internet** securely transfers the transaction.
- c. The bank's **computer network** verifies the account and completes the transfer.

Without the NII, online banking would not be possible.

Message distribution infrastructure

- This layer consists of software for sending and receiving messages.
- Its purpose is to **deliver a message from a server to a client**.
- For example, it could move an HTML file from a Web server to a client running Google Chrome. Messages can be unformatted (e.g., e-mail) or formatted (e.g., a purchase order).
- Electronic data interchange (EDI), e-mail, and hypertext text transfer protocol (HTTP) are examples of **messaging software**.

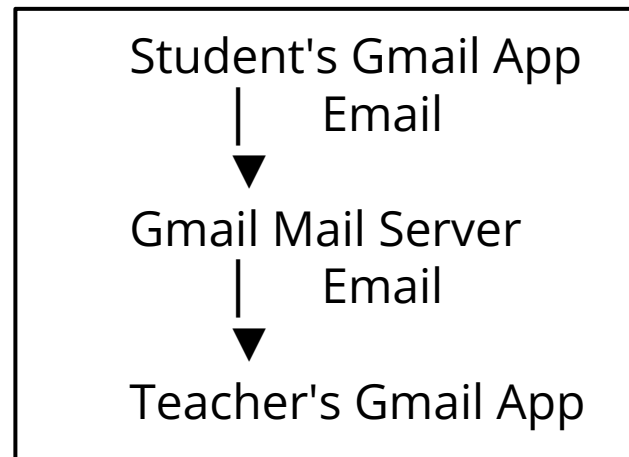
Message distribution infrastructure

- **Example 1: Sending an Email**

Scenario: A student sends an email to a teacher.

- Client:** Student's Email Application (e.g., Gmail)
- Server:** Gmail Mail Server
- Message Distribution Software:** E-mail Protocol (SMTP)
- Message:** Email

Flow of Message



Electronic publishing infrastructure

- Concerned with content, the Web is a very good example of this layer. It permits organizations to publish a full range of text and multimedia.
- There are three key elements of the Web:
 - A uniform resource locator (URL), which is used to uniquely identify any server;
 - A network protocol, enables communication between web browsers and servers.
 - A structured markup language, HTML, defines the structure and presentation of web pages.
- **Built on the NII (TCP/IP) layer:**
 - a. Uses **TCP/IP** for network communication.
 - b. Provides higher-level web services through **URL, HTTP, and HTML**.

Electronic publishing infrastructure

- **Example 1: Opening the Amazon Website**

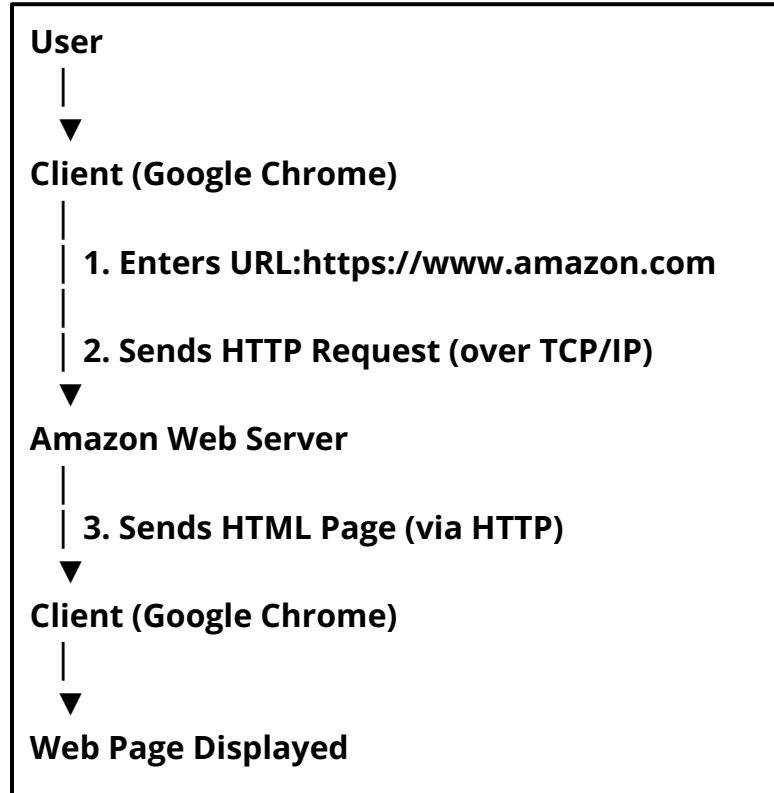
Scenario: A user visits **www.amazon.com** using Google Chrome.

- Client:** Google Chrome (Web Browser)
- Server:** Amazon Web Server
- URL:** <https://www.amazon.com>
- Network Protocol:** HTTP/HTTPS
- Markup Language:** HTML
- Underlying Network:** TCP/IP (NII Layer)

Example 1: Opening the Amazon Website

Component	Example
URL	https://www.amazon.com identifies the website.
HTTP	Transfers requests and responses between the browser and the server.
HTML	Defines the structure of the Amazon webpage.
TCP/IP	Carries all data across the Internet.

Flow of Message



Business services infrastructure

- The principal purpose of this layer is to support common business processes.
- Nearly every business is concerned with collecting payment for the goods and services it sells. Thus, the business services layer supports secure transmission of credit card numbers by providing encryption and electronic funds transfer.
- Furthermore, the business services layer should include facilities for encryption and authentication.

Business services infrastructure

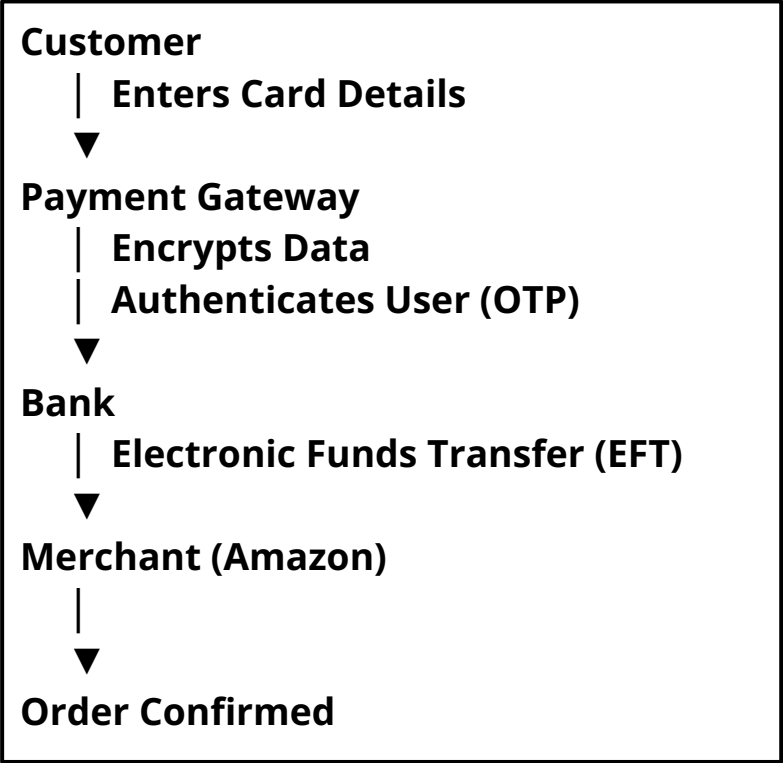
- **Example1: Online Shopping Payment**

Scenario: A customer buys a laptop from Amazon and pays using a credit/debit card.

- **Customer:** Enters card details on the checkout page.
- **Business Service:** Payment Gateway (e.g., Stripe, PayPal)
- **Security:** Card information is **encrypted** before transmission.
- **Authentication:** The bank verifies the customer using an **OTP** or **3D Secure**.
- **Electronic Funds Transfer (EFT):** The payment is transferred from the customer's bank account to the merchant's account.

Example1: Online Shopping Payment

Component	Example
Business Process	Online Payment
Encryption	Protects credit card information during transmission.
Authentication	OTP or 3D Secure verifies the customer.
Electronic Funds Transfer (EFT)	Transfers money from the customer's bank to the merchant's bank.



Electronic commerce applications

- Finally, on top of all the other layers sits an application.
- Consider the case of a book seller with an online catalog.
 - The application is a book catalog;
 - encryption is used to protect a customer's credit card number;
 - the application is written in HTML;
 - HTTP is the messaging protocol; and
 - the Internet physically transports messages between the book seller and customer.

Electronic commerce applications	Book catalog
Business services infrastructure	Encryption
Electronic publishing infrastructure	HTML
Message distribution infrastructure	HTTP
National information infrastructure	Internet

Fig: An electronic commerce application

Electronic publishing(E-publishing)

- Electronic Publishing (E-Publishing) means **creating, storing, and distributing digital content over electronic devices or the Internet instead of printing it on paper.**
- **Definition:** Electronic Publishing (E-Publishing) is the process of publishing and sharing information in digital form through the Internet or other electronic media.
- Two common approaches to electronic publishing are
 - Adobe's portable document format (**PDF**) and
 - **HTML.**

PDF

- PDF is a page description language that captures electronically the layout of the original document.
- Adobe's Acrobat Exchange software permits any document created by a DOS, Windows, or Unix application to be converted to PDF.
- Producing a PDF document is very similar to printing, except the image is sent to a file instead of a printer. The integrity of the original document is maintained—text, graphics, and tables are faithfully reproduced when the PDF file is printed or viewed.

PDF

- PDF is an operating system independent and printer independent way of presenting the same text and images on many different systems.
- PDF has been adopted by a number of organizations, including the Internal Revenue Service for tax forms.
- PDF documents can be sent as email attachments or accessed from a Web application.
- To decipher a PDF file, the recipient must use a special reader, supplied at no cost by Adobe for all major operating systems.

HTML

- HTML is a markup language , which means it marks a portion of text as referring to a particular type of information.
- HTML does not specify how this is to be interpreted; this is the function of the browser.
- Often the person using the browser can specify how the information will be presented.
- For instance, using the preference features of your browser, you can indicate the font and size for presenting information. As a result, you can significantly alter the look of the page, which could have been carefully crafted by a graphic artist to convey a particular look and feel. Thus, the you may see an image somewhat different from what the designer intended.

Difference between PDF & HTML

HTML

A markup language

HTML files can be created by a wide variety of software. Most word processors can generate HTML

Browser is free

Captures structure

Can have links to PDF

Reader can change presentation

PDF

A page description language

PDF files are created using special software sold by Adobe that is more expensive than many HTML creator alternatives

Viewer is free

Captures structure and layout

Can have links to HTML

Creator determines presentation

HTML or PDF?

- The choice between HTML and PDF depends on the main purpose of the document.
- If the intention is to inform the reader, then there is generally less concern with how the information is rendered. As long as the information is readable and presented clearly, the reader can be given control of how it is presented.

HTML or PDF?

- Alternatively, if the goal is to influence the reader (e.g., an advertisement) or maintain the original look of the source document (e.g, a taxation form or newspaper), then PDF is the better alternative.
- Also, a number of leading software companies are working on extensions to HTML that will give the creator greater control of the rendering of HTML. For example, they can specify the **font, font size, color, alignment, and layout** instead of relying on the browser's default settings.